SEMESTER - VII

COURSE: MAJOR – 16 (MJ-16) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: ELECTRODYNAMICS AND PLASMA PHYSICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to develop a deep understanding of classical and relativistic electrodynamics and their applications to radiating systems and plasma physics. It also equips students with tools to approach advanced problems in electrodynamics, radiation, and plasma phenomena, with relevance in applied physics contexts.

Learning Outcomes:

1. Analyze electromagnetic fields of moving charges via retarded and Liénard-Wiechert potentials.
2. Understand radiation from accelerated charges, dipoles, and antennas, focusing on energy distribution and angular dependence.
3. Explore relativistic electrodynamics with Lorentz transformations and Maxwell's equations covariance.
4. Learn plasma physics fundamentals: kinetic/fluid models, oscillations, shielding, and wave propagation in magnetized plasmas.
5. Apply concepts to linear antennas, antenna arrays, and plasma confinement devices.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Fields of moving charges and Radiating System (15 Lectures): Retarded Potentials, Lienard-Wiechert potentials, field of a point charge in uniform rectilinear motion, in arbitrary motion, Radiation from an accelerated charged particle at low and high velocity. Radiating System: Oscillating electric dipole, radiation from an oscillating dipole, from a small current element, from a linear antenna, Antenna arrays.

UNIT-II:

Relativistic Electrodynamics (15 Lectures): Transformation equation for current density and charge density, vector potential and scalar potentials, the electromagnetic field tensor, transformation equation for electric and magnetic field, Covariance of Maxwell equation in four tensor form, covariance of Maxwell and transformation law of Lorentz force.

UNIT-III:

Plasma Physics (15 Lectures): Elementary concepts of plasma, derivation of moment equations from Boltzmann equation. Plasma oscillation, Debye shielding, plasma confinement, magneto plasma. Fundamental equations, hydromagnetic waves: magnetosonic waves, Alfven waves, wave propagation parallel and perpendicular to magnetic field.

SUGGESTED READINGS:

1. Classical Electrodynamics, J.D. Jackson, 3rd Edn., 1998, Wiley.
2. The Classical Theory of Fields, L.D Landau, E.M Lifshitz, 4th Edn., 2003, Elsevier.
3. Introduction to Electrodynamics, D.J. Griffiths, 2012, Pearson Education.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Analyze the radiation field from a short current-carrying conductor.
2. Simulate and visualize the scalar and vector retarded potentials due to a point charge in uniform motion using computational methods.
3. Simulate the radiation fields of a time-harmonic electric dipole and observe the angular dependence of radiated power.

4. Simulate the force on a charged particle under transformed electric and magnetic fields in different inertial frames.
5. Simulate the oscillation of electrons about equilibrium position in plasma and determine the plasma frequency for various electron densities.
6. Model and analyze how a test charge is shielded in a plasma, and to calculate the Debye length for given plasma parameters.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VII

COURSE: MAJOR – 17 (MJ-17) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: MATHEMATICAL PHYSICS-II **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to equip students with a solid foundation in differential equations, vector calculus, complex analysis, Fourier series, Fourier transforms, and Laplace transforms, providing them with essential mathematical tools for solving problems in various scientific and engineering fields.

Learning Outcomes:

1. Analyze periodic functions and represent them using Fourier series, understanding orthogonality conditions, Dirichlet conditions, and complex representations of Fourier series.
2. Apply Fourier transforms to analyze and represent functions in the frequency domain, understanding properties such as convolution theorem and inverse Fourier transform.
3. Understand Laplace transforms, including their application to elementary functions, properties such as change of scale theorem and shifting theorem, and their use in solving differential equations and convolution problems.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Fourier Series (15 Lectures): Periodic functions, Orthogonality of sine and cosine functions, Dirichlet Conditions (Statement only), Expansion of periodic functions in a series of sine and cosine functions and

determination of Fourier coefficients, Complex representation of Fourier series, Analysis of saw tooth, triangular and square wave form.

UNIT-II:

Integral Transforms (15 Lectures): Fourier transform of trigonometric, Gaussian, finite wave train & other functions, Representation of Dirac delta function as a Fourier Integral, Fourier transform of derivatives, Inverse Fourier transform, Convolution theorem. Laplace Transform (LT) of Elementary functions, Properties of LTs: Change of Scale Theorem, Shifting Theorem, Dirac Delta function, Periodic Functions, Convolution Theorem.LT of derivatives and integral, Inverse LT and applications of LT.

UNIT-III:

Frobenius Method and Special Functions (15 Lectures):: Singular points of second order linear differential equations and their importance, Frobenius method and its applications to differential equations, Legendre, Bessel, Hermite and Laguerre functions, properties and differential equations, generating function, recurrence relation, orthogonality.

SUGGESTED READINGS:

1. Mathematical Methods for Physicists, G.B. Arfken, H.J. Weber, F.E. Harris, 2013, 7th Edn., Elsevier.
2. Mathematical Physics, H. K. Dass and Rama Verma, S. Chand & Company Pvt. Ltd.
3. Modern Mathematical Methods for Physicists and Engineers, C.D. Cantrell, 2011, Cambridge University Press.
4. Mathematical Methods for Physics & Engineers, K. F. Riley, M. P. Hobson, S. J. Bence, 3rd Ed., 2006, Cambridge University Press.
5. Mathematical physics by B.S. Rajput.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Verify the orthogonality of sine and cosine functions over a given interval.
2. Compute and visualize the Fourier transforms of standard functions like sine, cosine, Gaussian, and finite wave train functions.

3. Demonstrate the representation of the Dirac delta function as the limiting case of a Fourier integral using a numerical approach.
4. Verify the properties of Laplace transforms: linearity, shifting, scaling, and the convolution theorem.
5. Plot Legendre polynomials for various orders.
6. Compute Bessel functions of the first kind and verify recurrence relations and orthogonality.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VII

COURSE: MAJOR – 18 (MJ-18) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: QUANTUM MECHANICS-II **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to develop the advanced understanding about mathematical foundation useful to explain the various quantum phenomena such as scattering, perturbation theory and approximation methods.

Learning Outcomes:

1. Understand linear vector spaces, Dirac notation, and operators in quantum mechanics.
2. Analyze angular momentum operators, spin, and related symmetries.
3. Apply quantum scattering theory including partial waves and Born approximation.
4. Utilize time-independent and time-dependent perturbation theories.
5. Interpret physical phenomena such as Stark effect and transition probabilities.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Mathematical Foundation of Quantum Mechanics (15 Lectures): Vectors and Linear vector space, Closure property, Linear independence of vectors, Bases and dimensions. Some examples of linear vector spaces, Dirac's notations, Bra and Ket vectors, Combining bras with kets, Inner product and inner product space, Orthonormality of vectors, Completeness condition, Outer product, Hilbert space and the Coordinate Representation, Operator on a linear vector space, Algebra of linear operators.

UNIT-II:

Angular Momentum (15 Lectures): Commutation relations for angular momentum operators, Eigenvalues and eigenvectors, Pauli spin matrices and spin eigenvectors, Motion in a centrally symmetric field, Clebsch-Gordon Coefficients. Space-time symmetries and conservation Laws for linear momentum, Angular momentum, Energy and Parity. Electron Spin, Exclusion principle, Symmetric and Antisymmetric Wave Functions, Spin-Orbit Coupling, Total Angular Momentum.

UNIT-III:

Perturbation Theory and Scattering(15 Lectures): Time Independent Perturbation Theory (First and Second order), Degenerate and non-degenerate cases, Stark Effect, Time dependent perturbation theory, Transition Probability, Fermi's Golden Rule, Constant and Harmonic perturbation. Differential Scattering Cross Section and Total Scattering Cross Section, Formulation of quantum scattering theory, Scattering amplitude, Scattering of a Wave Packet, Green's function in Scattering theory, Partial wave analysis, Scattering in a Coulomb Field, Optical theorem, Born Approximation, Phase Shifts, Scattering length and effective range for short range potential.

SUGGESTED READINGS:

1. A Text Book of Quantum Mechanics, Mathews, P.M., &Venkatesan, K., TMH.
2. Quantum Mechanics, Merzbacker, E., John Wiley
3. Quantum Mechanics Messiah, A., North-Holland Publishing Co.
4. Quantum Mechanics Schiff, L.I., Tata McGraw-Hill, 3rd Edition 2010
5. Quantum Mechanics Ghatak, A., Narosa Publishing House, New Delhi.
6. Quantum Mechanics, Agarwal, B. K., PHI
7. Modern Quantum Mechanics, J.J. Sakurai
8. Quantum Mechanics, Landau, L.D. &Lifshitz, E.M., Pergman Press
9. Quantum Mechanics Concepts and Applications 2nd Edition NouredineZettili.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Verify the closure property and linear independence of a set of vectors in a finite-dimensional vector space using computational methods.

2. Compute inner products of vectors and verify orthonormality in an inner product space numerically.
3. Compute and visualize the differential and total scattering cross section for given potentials.
4. Simulate the scattering of a Gaussian wave packet from a one-dimensional potential and study reflection/transmission coefficients.
5. Compute and visualize the energy level splitting due to an external electric field (Stark effect) using perturbation theory.
6. Simulate the time evolution of quantum states under a time-dependent potential and compute transition probabilities.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VII

COURSE: ADVANCE MAJOR – 1(AMJ-1) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: ADVANCED QUANTUM MECHANICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to introduce approximation methods, relativistic wave equations, identical particles, and quantum entanglement. It also aims to develop the concepts in quantum field theory, including field quantization and gauge invariance, preparing students for research in modern theoretical physics and applications in atomic, particle, and quantum information science.

Learning Outcomes:

1. Apply approximation methods like variation principle and WKB to analyze non-exactly solvable quantum systems.
2. Understand and solve relativistic quantum equations such as the Klein-Gordon and Dirac equations, and interpret their physical implications.
3. Distinguish between bosons and fermions using exchange symmetry, and utilize second quantization and Fock space formalism.
4. Analyze quantum entanglement using reduced density matrices, entanglement entropy, and Bell's inequalities.
5. Comprehend basic principles of quantum field theory, including field quantization, propagators, Feynman diagrams, and gauge invariance.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Approximation methods (10 Lectures): Variation methods and its application to Ground State of Hydrogen Atom and First excited state of Harmonic oscillator, WKB approximation, Interaction with classical electromagnetic fields,

UNIT-II:

Relativistic Quantum Mechanics (15 Lectures): Klein-Gordon (K-G) equation, Limitations of K-G equations, Dirac equation and matrices, Plain wave solution of Dirac equation, Probability density and current density, Concept of negative energy state of electron. Central field problem: exact solution, Hydrogen atom and hyperfine structure, Discrete symmetries: parity, charge reversal, time reversal.

UNIT-III:

Identical Particles and Quantum Field Theory (20 Lectures): Exchange symmetry: Bosons and Fermions, Second quantization formalism, Fock space and creation/annihilation operators, Entanglement and Bell's inequalities, Reduced density matrices and quantum entropy, Entanglement entropy. Quantization of scalar and spinor fields, Canonical quantization and path integral quantization, Normal ordering and vacuum energy, Propagators and Feynman diagrams, Gauge invariance.

SUGGESTED READINGS:

1. A Text Book of Quantum Mechanics, Mathews, P.M., &Venkatesan, K., TMH.
2. Quantum Mechanics, Merzbacker, E., John Wiley
3. Quantum Mechanics Messiah, A., North-Holland Publishing Co.
4. Quantum Mechanics Schiff, L.I., Tata McGraw-Hill, 3rd Edition 2010
5. Quantum Mechanics Ghatak, A., Narosa Publishing House, New Delhi.
6. Quantum Mechanics, Agarwal, B. K., PHI
7. Modern Quantum Mechanics, J.J. Sakurai
8. Quantum Mechanics, Landau, L.D. &Lifshitz, E.M., Pergman Press

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Compute the energy of the first excited state of a one-dimensional quantum harmonic oscillator and compare it with the exact solution.
2. Numerically solve the Klein-Gordon equation for a free particle and study the behavior of its solutions.
3. Simulate the behavior of two identical particles (bosons and fermions) and verify the exchange symmetry of their wavefunctions.
4. Compute the positive and negative energy solutions of the Dirac equation and visualize their spinor components.
5. Numerically evaluate the path integral for a free particle and a harmonic oscillator.
6. Simulate entangled states and verify the violation of Bell's inequality.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - VII

COURSE: RESEARCH METHODOLOGY – 1(RM-1) **TOTAL CREDITS:** THEORY-04
PAPER NAME: RESEARCH METHODOLOGY **TEACHING HOURS:** THEORY-60

EVALUATION		
	External Exam	**Internal Exam**
Full Marks	**75**	25 (20 Written + 5 Attendance/Overall Class Performance)
Duration of Exam	3 Hours	1 Hour
Pass Marks	40 Marks	

Course Objective:

This course aims to inculcate research aptitude among the learners and equip them with knowledge and skills required to successfully undertake various steps in the research process.

Learning Outcomes:

1. Able to analyze research concepts, its types and steps in the research process.
2. Identify the appropriate research problems and approach to solve them.
3. Recognize various techniques of data analysis and interpretation, and learn different mathematical modeling.
4. Develop the sense of ethics in research process.
5. Able to prepare a complete research report in appropriate format and searching of reputed journals for publishing their articles/reports.

Course Content:

Introduction to Science and Scientific Thinking (Lectures: 10): What is science? Characteristics of scientific knowledge, Objectivity vs subjectivity, scientific reasoning: Inductive and deductive logic, Examples of scientific discoveries and logical inference.

Scientific Method, Hypothesis and scientific communication (Lectures: 15): How scientists ask questions, Hypotheses: definition, examples, testability, Simple data collection and experimental ideas, Understanding variables, observations, and inference**.** Basics of Scientific Communication: Writing lab reports and project summaries, Abstract, introduction, and conclusion, Preparing posters and visual aids, Basics of citation and plagiarism.

Research Ethics, Scientific Attitude, Scientific Database and Publications (Lectures: 25): Scientific honesty and integrity, what is plagiarism? How to avoid it? Importance of accurate data recording and openness in research, Role of collaboration and communication in research. **Scientific Database and Publications:** UGC Care, Scopus, Google Scholar, Web of Science, Science Citation Index (SCI), Orchid, Archives. Types of Publications: Articles, Books, Journals, Archives, Conference Papers, Theses, Patents, Journal Impact: Journal citation reports, SCImago Journal Rank (SJR), Cite score, Source Normalized Impact per Paper (SNIP), h-index.

Note: Internal assessment will include assignments, short project reports, and classroom participation.

SUGGESTED READINGS:

1. Soumitro Banerjee, Research Methodology for Natural Sciences, IISc Press, 2022.
2. Research Methodology- C. R. Kothari

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